Platonic Solids

An attempt

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Introduction to the 5 Platonic solids

Platonic solids are a special class of highly symmetric three-dimensional geometric objects. They are named after the Greek philosopher Plato, who discussed them in his work *Timaeus* and associated them with the classical elements of nature. However, the mathematical study of these solids predates Plato and was known to earlier Greek mathematicians.

A Platonic solid is a **regular convex polyhedron** satisfying the following conditions:

- All faces are **congruent regular polygons**.
- The same number of faces meet at every vertex.
- All edges are equal in length.
- All vertices are geometrically identical.

Because of these strict conditions, only **five** Platonic solids exist.

Platonic Solid	Faces	Face Type	Vertices	Edges	Hamiltonian	Eulerian
Tetrahedron	4	Equilateral Triangle	4	6	Yes	No
Cube (Hexahedron)	6	Square	8	12	Yes	No
Octahedron	8	Equilateral Triangle	6	12	Yes	Yes
Dodecahedron	12	Regular Pentagon	20	30	Yes	No

Icosahedron	20	Equilateral Triangle	12	30	Yes	No

The existence of only five Platonic solids can be understood through geometric constraints. At each vertex, the sum of the face angles meeting there must be **less than** (360°); otherwise, the shape would lie flat and fail to form a three-dimensional solid. This restriction allows only a limited number of regular polygons to combine in space, resulting in exactly five possible solids.

The Platonic solids possess remarkable symmetry and play an important role in geometry, topology, crystallography, architecture, chemistry, and mathematical modelling. Several important mathematical concepts such as duality, symmetry groups, Euler's formula, and polyhedral geometry can be explored through their study.

The five Platonic solids are:

1. **Tetrahedron** — composed of four equilateral triangular faces.
2. **Cube (Hexahedron)** — composed of six square faces.
3. **Octahedron** — composed of eight equilateral triangular faces.
4. **Dodecahedron** — composed of twelve regular pentagonal faces.
5. **Icosahedron** — composed of twenty equilateral triangular faces.

These solids are closely related through the concept of **duality**. The tetrahedron is self-dual, while the cube and octahedron form a dual pair, as do the dodecahedron and icosahedron.

The study of Platonic solids provides a foundation for understanding regular polyhedra and offers insight into the deep relationship between symmetry, geometry, and spatial structure.

Tetrahedron

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Geometric Properties

Let the edge length be a.

Property	Formula / Value
Faces	4
Edges	6
Vertices	4
Face Type	Equilateral Triangle

Property	Formula / Value
Face Interior Angle	60°
Dihedral Angle	$\cos^{-1}\left(\frac{1}{3}\right) \approx 70.53^\circ$
Surface Area	$\sqrt{3}a^2$
Volume	$\frac{\sqrt{2}}{12}a^3$
Height	$\sqrt{\frac{2}{3}}a$
Inradius (r)	$\frac{\sqrt{6}}{12}a$
Circumradius (R)	$\frac{\sqrt{6}}{4}a$
Midradius (ρ)	$\frac{\sqrt{2}}{4}a$

Spectral Properties

Adjacency Matrix

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

$\text{Spec}(A) = \{(3)^1, (-1)^3\}$

Laplacian Matrix

$$L = \begin{bmatrix} 3 & -1 & -1 & -1 \\ -1 & 3 & -1 & -1 \\ -1 & -1 & 3 & -1 \\ -1 & -1 & -1 & 3 \end{bmatrix}$$

$\text{Spec}(L) = \{(4)^3, (0)^1\}$

Singular Values of Incidence Matrix

$\sigma(Q) = \{(2)^3, (0)^1\}$

Symmetry and Automorphisms

For Platonic solids, the total number of geometric symmetries of the solid is equal to the total number of automorphisms of its associated graph. In other words, every graph automorphism corresponds to a geometric symmetry, and every geometric symmetry induces a graph automorphism.

$|Sym(Solid)| = |Aut(G)|$

The total number of symmetries is calculated by counting all distinct rotations and reflections about the axes of symmetry passing through vertices, edges, and faces of the solid. These symmetries together form the symmetry group of the Platonic solid.

To count:

Identity = 1

Vertex to centre of opposite face -> 4 axes and each face is a triangle -> 2 rotations per axis $(120°, 240°) = 4 \times 2 = 8$

Edge to opposite edge -> 3 axes and total rotations per axis -> 1 $(180°) = 3 \times 1$ $= 3$

Total rotations = 1+8+3 = 12

$|Rot(Tetrahedron)| = 12$

All rotational symmetries of the tetrahedron correspond to even permutations of the 4 vertices.

$Rot(Tetrahedron) \cong A_4$

Total reflections = Total rotations = 12

Total symmetries = total rotations + total reflections = 12+12 = 24

$|Sym(Tetrahedron)| = 24$

$Sym(Tetrahedron) \cong S_4$

Cube

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Geometric Properties

Let the edge length be a.

Property	Formula / Value
Faces	6
Edges	12
Vertices	8
Face Type	Square
Face Interior Angle	90°
Dihedral Angle	90°
Surface Area	$6a^2$
Volume	a^3
Space Diagonal	$\sqrt{3}a$
Face Diagonal	$\sqrt{2}a$
Inradius (r)	$\frac{a}{2}$
Circumradius (R)	$\frac{\sqrt{3}}{2}a$
Midradius (ρ)	$\frac{\sqrt{2}}{2}a$

Spectral Properties

Adjacency Matrix

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \end{bmatrix}$$

$\text{Spec}(A) = \{(3)^1, (1)^3, (-1)^3, (-3)^1\}$

Laplacian Matrix

$$L = \begin{bmatrix} 3 & -1 & 0 & -1 & -1 & 0 & 0 & 0 \\ -1 & 3 & -1 & 0 & 0 & -1 & 0 & 0 \\ 0 & -1 & 3 & -1 & 0 & 0 & -1 & 0 \\ -1 & 0 & -1 & 3 & 0 & 0 & 0 & -1 \\ -1 & 0 & 0 & 0 & 3 & -1 & 0 & -1 \\ 0 & -1 & 0 & 0 & -1 & 3 & -1 & 0 \\ 0 & 0 & -1 & 0 & 0 & -1 & 3 & -1 \\ 0 & 0 & 0 & -1 & -1 & 0 & -1 & 3 \end{bmatrix}$$

$\text{Spec}(L) = \{(6)^1, (4)^3, (2)^3, (0)^1\}$

Singular Values of Incidence Matrix

$\sigma(Q) = \{(\sqrt{6})^1, (2)^3, (\sqrt{2})^3, (0)^1\}$

Symmetries and Automorphisms

To count:

Identity = 1

Centre of opposite faces -> 3 axes and each face is a square -> 3 rotations per axis $(90^\circ, 180^\circ, 270^\circ) = 3 \times 3 = 9$

Vertex to opposite vertex -> 4 axes and each vertex has degree 3 -> 2 rotations per axis $(120^\circ, 240^\circ) = 4 \times 2 = 8$

Midpoint of opposite edges -> 6 axes and total rotations per axis -> 1 $(180^\circ) = 6 \times 1 = 6$

Total rotations = 1+9+8+6 = 24

$|Rot(Cube)| = 24$

All rotations of the cube correspond to permutations of the 4 body diagonals.

$Rot(Cube) \cong S_4$

Total reflections = Total rotations = 24

Total symmetries = total rotations + total reflections = 24+24 = 48

$|Sym(Cube)| = 48$

$Sym(Cube) \cong S_4 \times C_2$

Octahedron

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Geometric Properties

Let the edge length be a.

Property	Formula / Value
Faces	8
Edges	12
Vertices	6
Face Type	Equilateral Triangle
Face Interior Angle	60°
Dihedral Angle	$\cos^{-1}\left(-\frac{1}{3}\right) \approx 109.47^\circ$
Surface Area	$2\sqrt{3}\,a^2$
Volume	$\frac{\sqrt{2}}{3}a^3$
Height (vertex-to-vertex)	$\sqrt{2}\,a$
Inradius (r)	$\frac{\sqrt{6}}{6}a$
Circumradius (R)	$\frac{\sqrt{2}}{2}a$
Midradius (ρ)	$\frac{1}{2}a$

Spectral Properties

Adjacency Matrix

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 0 \end{bmatrix}$$

$\text{Spec}(A) = \{(4)^1, (0)^3, (-2)^2\}$

Laplacian Matrix

$$L = \begin{bmatrix} 4 & -1 & -1 & -1 & -1 & 0 \\ -1 & 4 & -1 & 0 & -1 & -1 \\ -1 & -1 & 4 & -1 & 0 & -1 \\ -1 & 0 & -1 & 4 & -1 & -1 \\ -1 & -1 & 0 & -1 & 4 & -1 \\ 0 & -1 & -1 & -1 & -1 & 4 \end{bmatrix}$$

$\mathrm{Spec}(L) = \{(6)^2, (4)^3, (0)^1\}$

Singular Values of Incidence Matrix

$\sigma(Q) = \{(\sqrt{6})^2, (2)^3, (0)^1\}$

Symmetries and Automorphisms

To count:

Identity = 1

Centre of opposite faces -> 4 axes and each face is a triangle -> 2 rotations per axis $(120^\circ, 240^\circ) = 4 \times 2 = 8$

Vertex to opposite vertex -> 3 axes and each vertex has degree 4 -> 3 rotations per axis $(90^\circ, 180^\circ, 270^\circ) = 3 \times 3 = 9$

Midpoint of opposite edges -> 6 axes and total rotations per axis -> 1 $(180^\circ) = 6 \times 1 = 6$

Total rotations = 1+8+9+6 = 24

$|Rot(Octahedron)| = 24$

All rotational symmetries of the octahedron correspond to permutations of the 4 pairs of opposite faces.

$Rot(Octahedron) \cong S_4$

Total reflections = Total rotations = 24

Total symmetries = total rotations + total reflections = 24+24 = 48

$|Sym(Octahedron)| = 48$

$Sym(Octahedron) \cong S_4 \times C_2$

Dodecahedron

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Geometric Properties

Let the edge length be a.

Define the golden ratio:

$$\phi = \frac{1+\sqrt{5}}{2}$$

Property	Formula / Value
Faces	12
Edges	30
Vertices	20
Face Type	Regular Pentagon
Face Interior Angle	108°
Dihedral Angle	$\cos^{-1}\left(-\frac{1}{\sqrt{5}}\right) \approx 116.57^\circ$
Surface Area	$3\sqrt{25+10\sqrt{5}}\,a^2$
Volume	$\frac{15+7\sqrt{5}}{4}a^3$
Pentagon Diagonal	ϕa
Inradius (r)	$\frac{a}{20}\sqrt{250+110\sqrt{5}}$
Circumradius (R)	$\frac{\sqrt{3}}{2}\phi a$
Midradius (ρ)	$\frac{1}{4}(3+\sqrt{5})a$

Spectral Properties

Adjacency Matrix

$$A = \begin{bmatrix}
0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
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0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 \\
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0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0
\end{bmatrix}$$

$\mathrm{Spec}(A) = \{(3)^1, (\sqrt{5})^3, (1)^5, (0)^4, (-2)^4, (-\sqrt{5})^3\}$

Laplacian Matrix

$$L = \left[\begin{array}{cccccccccccccccccccc}
3 & -1 & 0 & 0 & -1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-1 & 3 & -1 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -1 & 3 & -1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -1 & 3 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-1 & 0 & 0 & -1 & 3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\
-1 & 0 & 0 & 0 & 0 & 3 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -1 & 3 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\
0 & -1 & 0 & 0 & 0 & 0 & -1 & 3 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 3 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\
0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & -1 & 3 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 3 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\
0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 3 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 3 & -1 & 0 & 0 & 0 & 0 & 0 & -1 \\
0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 3 & -1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 3 & -1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 3 & -1 & 0 & 0 & -1 \\
0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 3 & -1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 3 & -1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 3 & -1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & -1 & 0 & 0 & -1 & 3
\end{array}\right]$$

$\mathrm{Spec}(L) = \{(3+\sqrt{5})^3, (5)^4, (3)^4, (2)^5, (3-\sqrt{5})^3, (0)^1\}$

Singular Values of Incidence Matrix

$\sigma(Q) = \{(\sqrt{3+\sqrt{5}})^3, (\sqrt{5})^4, (\sqrt{3})^4, (\sqrt{2})^5, (\sqrt{3-\sqrt{5}})^3, (0)^1\}$

Symmetries and Automorphisms

To count:

Identity = 1

Centre of opposite faces -> 6 axes and each face is a pentagon -> 4 rotations per axis $(72^\circ, 144^\circ, 216^\circ, 288^\circ) = 6 \times 4 = 24$

Vertex to opposite vertex -> 10 axes and each vertex has degree 3 -> 2 rotations per axis $(120^\circ, 240^\circ) = 10 \times 2 = 20$

Midpoint of opposite edges -> 15 axes and total rotations per axis -> 1 $(180^\circ) = 15 \times 1 = 15$

Total rotations = 1+24+20+15 = 60

$|Rot(Dodecahedron)| = 60$

All rotational symmetries of the dodecahedron correspond to even permutations of the 5 inscribed cubes.

$Rot(Dodecahedron) \cong A_5$

Total reflections = Total rotations = 60

Total symmetries = total rotations + total reflections = 60+60 = 120

$|Sym(Dodecahedron)| = 120$

$Sym(Dodecahedron) \cong A_5 \times C_2$

Icosahedron

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Geometric Properties

Let the edge length be a.

Define the golden ratio:

$$\phi = \frac{1+\sqrt{5}}{2}$$

Property	Formula / Value
Faces	20
Edges	30
Vertices	12
Face Type	Equilateral Triangle
Face Interior Angle	60°
Dihedral Angle	$\cos^{-1}\left(-\frac{\sqrt{5}}{3}\right) \approx 138.19^\circ$
Surface Area	$5\sqrt{3}\,a^2$
Volume	$\frac{5(3+\sqrt{5})}{12}a^3$
Height (vertex-to-vertex)	$\phi\sqrt{5}\,a$
Inradius (r)	$\frac{\sqrt{3}}{12}(3+\sqrt{5})a$
Circumradius (R)	$\frac{1}{4}\sqrt{10+2\sqrt{5}}\,a$
Midradius (ρ)	$\frac{\phi^2}{2}a$

Spectral Properties

Adjacency Matrix

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 \end{bmatrix}$$

$\mathrm{Spec}(A) = \{(5)^1, (\sqrt{5})^3, (-1)^5, (-\sqrt{5})^3\}$

Laplacian Matrix

$$L = \begin{bmatrix} 5 & -1 & -1 & -1 & -1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ -1 & 5 & -1 & 0 & -1 & -1 & -1 & 0 & 0 & 0 & 0 & 0 \\ -1 & -1 & 5 & 0 & 0 & 0 & -1 & -1 & -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 5 & -1 & 0 & 0 & 0 & -1 & -1 & -1 & 0 \\ -1 & -1 & 0 & -1 & 5 & -1 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 & -1 & 5 & -1 & 0 & 0 & 0 & -1 & -1 \\ 0 & -1 & -1 & 0 & 0 & -1 & 5 & -1 & 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 & 0 & 0 & -1 & 5 & -1 & -1 & 0 & -1 \\ -1 & 0 & -1 & -1 & 0 & 0 & 0 & -1 & 5 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & -1 & -1 & 5 & -1 & -1 \\ 0 & 0 & 0 & -1 & -1 & -1 & 0 & 0 & 0 & -1 & 5 & -1 \\ 0 & 0 & 0 & 0 & 0 & -1 & -1 & -1 & 0 & -1 & -1 & 5 \end{bmatrix}$$

$\mathrm{Spec}(L) = \{(5+\sqrt{5})^3, (6)^5, (5-\sqrt{5})^3, (0)^1\}$

Singular Values of Incidence Matrix

$\sigma(Q) = \{(\sqrt{5+\sqrt{5}})^3, (\sqrt{6})^5, (\sqrt{5-\sqrt{5}})^3, (0)^1\}$

Symmetries and Automorphisms

To count:

Identity = 1

Vertex to opposite vertex -> 6 axes and each vertex has degree (5) -> 4 rotations per axis $(72^\circ, 144^\circ, 216^\circ, 288^\circ) = 6 \times 4 = 24$

Centre of opposite faces -> 10 axes and each face is a triangle -> 2 rotations per axis $(120^\circ, 240^\circ) = 10 \times 2 = 20$

Midpoint of opposite edges -> 15 axes and total rotations per axis -> 1 $(180^\circ) = 15 \times 1 = 15$

Total rotations = 1+24+20+15 = 60

$|Rot(Icosahedron)| = 60$

All rotational symmetries of the icosahedron correspond to even permutations of the 5 cubes dual to the inscribed dodecahedral structure.

$Rot(Icosahedron) \cong A_5$

Total reflections = Total rotations = 60

Total symmetries = total rotations + total reflections = 60+60 = 120

$|Sym(Icosahedron)| = 120$

$Sym(Icosahedron) \cong A_5 \times C_2$

What makes a solid Platonic?

Definition and proof

A solid is called **Platonic** if it is a **convex polyhedron with congruent regular polygonal faces** and possesses complete symmetry through **flag-transitivity**.

A *flag* consists of a vertex, an incident edge, and an incident face. A polyhedron is said to be **flag-transitive** if any flag can be mapped to any other flag by a symmetry (rotation or reflection) of the solid. This ensures that no face, edge, vertex, or local configuration is geometrically distinguished from another.

Consequently, a Platonic solid satisfies the following equivalent properties:

- All faces are congruent regular polygons.
- The same number of faces meet at every vertex.
- All edges have equal length.
- All vertices, edges, and faces are equivalent under symmetry.

These conditions formalize the idea of a **perfectly regular three-dimensional solid**. Convexity prevents self-intersections, regular faces ensure uniformity, and flag-transitivity guarantees complete geometric symmetry.

To determine all possible Platonic solids, let each face be a regular p-gon and let q faces meet at every vertex. A Platonic solid is denoted by its **Schläfli symbol** $\{p, q\}$, where:

- p represents the number of edges (or sides) of each face,
- q represents the number of faces meeting at each vertex.

For a convex polyhedron to exist, the total angle around a vertex must be strictly less than 360°,

$q\left(\frac{(p-2)180^\circ}{p}\right) < 360^\circ$

which simplifies to

$\frac{1}{p} + \frac{1}{q} > \frac{1}{2}$

The only integer solutions with $p, q \geq 3$ are

$\{3,3\},\ \{4,3\},\ \{3,4\},\ \{5,3\},\ \{3,5\}$

corresponding respectively to:

- $\{3,3\}$ — Tetrahedron
- $\{4,3\}$ — Cube
- $\{3,4\}$ — Octahedron
- $\{5,3\}$ — Dodecahedron
- $\{3,5\}$ — Icosahedron

Hence, exactly **five Platonic solids** exist.

Relaxing the Conditions of Platonic Solids

Platonic solids arise from three fundamental conditions:

1. **Convexity** — the solid has no self-intersections or inward indentations.
2. **Regular polygonal faces** — every face is a congruent regular polygon.
3. **Flag-transitivity** — any flag (a vertex, an incident edge, and an incident face) can be mapped to any other flag by a symmetry of the solid.

These conditions are highly restrictive and uniquely determine the five Platonic solids. However, if one of the conditions is relaxed, new families of polyhedra emerge.

Removing Convexity

If convexity is removed while preserving regular polygonal faces and flag-transitivity, one obtains the **Kepler–Poinsot solids**. These are non-convex regular star polyhedra that retain complete symmetry but allow self-intersecting geometry.

Removing Flag-Transitivity

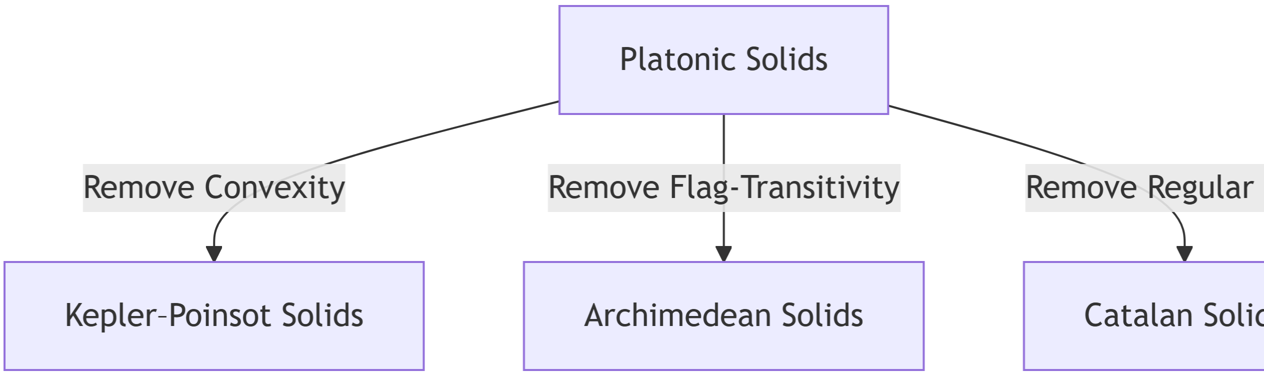
If flag-transitivity is removed while preserving convexity and regular polygonal faces, one obtains the **Archimedean solids**. These solids remain highly symmetric and vertex-transitive, but different face types may occur, preventing complete flag equivalence.

For example, the cuboctahedron contains both triangular and square faces, so a symmetry cannot map a flag involving a triangle to one involving a square.

Removing Regular Polygonal Faces

If regular polygonal faces are relaxed while preserving convexity and flag-transitivity, one obtains the **Catalan solids**, which are the duals of the Archimedean solids. These solids preserve strong symmetry, although their faces are no longer regular polygons.

Thus, the Platonic solids may be viewed as the **most restrictive and maximally symmetric class of convex polyhedra**, arising only when all three defining conditions are simultaneously satisfied.



Consequences of a Solid Being Platonic

Once a solid satisfies the defining conditions of Platonicity, several important geometric, combinatorial, and algebraic properties follow naturally.

Symmetry and Transitivity

Platonic solids exhibit the greatest degree of symmetry among convex polyhedra. Their symmetry groups act transitively on flags, implying that every incident vertex-edge-face triple can be mapped to any other through a symmetry of the solid.

Consequently, Platonic solids are:

- **vertex-transitive**, meaning every vertex is structurally identical,
- **edge-transitive**, meaning every edge occupies an equivalent position,
- **face-transitive**, meaning every face is symmetrically equivalent.

Thus, no vertex, edge, or face is distinguished from another, and every local region of the solid behaves identically.

This complete regularity distinguishes Platonic solids from other polyhedral families. For example:

- **Archimedean solids** are vertex-transitive but generally fail to be face-transitive, since they contain multiple types of regular polygonal faces.
- **Catalan solids** are face-transitive but fail to be vertex-transitive, since vertices may possess different local configurations.

The rotational symmetry groups of the Platonic solids are:

Platonic Solid	Rotational Symmetry Group
Tetrahedron	A_4
Cube	S_4
Octahedron	S_4
Dodecahedron	A_5
Icosahedron	A_5

These symmetry groups explain the strong uniformity observed in Platonic solids and contribute significantly to their structured spectral properties.

Regular Graph Structure

The graph associated with a Platonic solid is a **regular graph**.

If a Platonic solid has Schläfli symbol $\{p, q\}$, then every face is a p-gon and exactly q faces meet at each vertex. Hence, every vertex has degree q, making the adjacency graph q-regular.

For a q-regular graph,

$$A\mathbf{1} = q\mathbf{1}$$

where A is the adjacency matrix and $\mathbf{1}$ is the all-ones vector. Consequently, q is always an eigenvalue of the adjacency matrix.

Duality

Every Platonic solid possesses a Platonic dual.

The duality relationships are

$$\text{Cube} \leftrightarrow \text{Octahedron}$$

$$\text{Dodecahedron} \leftrightarrow \text{Icosahedron}$$

while the tetrahedron is **self-dual**.

In the dual polyhedron, vertices and faces interchange roles while preserving regularity.

Spectral Properties

The regularity and symmetry of Platonic solids strongly influence the spectra of their adjacency and Laplacian matrices.

Some immediate consequences include:

- the largest adjacency eigenvalue equals the vertex degree,
- repeated eigenvalues arising from symmetry,
- highly structured eigenspaces,
- spectral patterns reflecting geometric regularity.

These spectral properties provide insight into connectivity, combinatorial structure, and symmetry, making spectral graph theory an effective tool for studying Platonic solids.